

Banda Karthik

Bachelor of Technology
Artificial Intelligence and Data Science
Lakireddy Bali Reddy College of Engineering,
Mylavaram

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Education

Degree/Certificate	Institute/Board	Percentage	Year
B.Tech., AI&DS	Lakireddy Bali Reddy College of Engineering	85.5%	Pursuing 2027
Intermediate	Sarada Educational Institutions	93.9%	2023
S.S.C	Sri Vijnana Vihara EM High School	100%	2021

Internships

- **Indian Institute of Technology [🌐]** *May 2026 - July 2026*
  Research Intern Guwahati, Assam
  - Contributed to an MPNN-based GNN pipeline predicting DFT energies of ZrO_2 crystal alloys
  - Parsed geometry files with regex and built atom-bond graphs with atomic and RBF edge features
  - Supported evaluation (R^2, MAE, RMSE); final model reached $R^2 = 0.974$ vs 0.789 baseline
  - Helped enable millisecond predictions replacing hours-long Quantum ESPRESSO calculations
- **Mindenious Edutech [🌐]** *May 2025 - July 2025*
  Java Intern Remote
  - Developed a Core Java elevator simulation handling multi-floor requests (floors 0–10)
  - Implemented SCAN-based scheduling using a `TreeSet` for sorted, duplicate-free requests
  - Applied encapsulation, enum-based direction states and `InterruptedException` handling

Projects

- **ChatzApplication: Real-Time One-to-One Chat Application** *July 2026*
  Tools: Java, Spring Boot, Hibernate, Oracle DB, WebSocket, STOMP, Maven, HTML, CSS, JavaScript [🐙]
  - Developed a full stack chat application with registration, login features.
  - Implemented real-time messaging with WebSocket and STOMP protocol
  - Built a layered Spring Boot backend (Controller, Service, Repository) with three Oracle tables (`USERS`, `MESSAGE`, `ONLINE_USER`) mapped through Hibernate
- **Al-Barakah: Responsive Food Ordering Website** *[March 2026]*
  Tools: HTML, CSS, JavaScript, localStorage, Service Workers, PWA, GitHub Pages [🐙]
  - Developed a responsive frontend-only ordering website for a fruit bowl startup, deployed on GitHub Pages
  - Implemented persistent cart management and zone-based delivery pricing with WhatsApp order integration
  - Built PWA support with a service worker for offline browsing.

Skills

- **Programming Languages**: Java, Python, C.
- **Web Technologies**: HTML, CSS, JavaScript, Spring, Spring Boot, Maven, Hibernate.
- **Databases**: SQL(MySQL)
- **Tools**:Git, VSCode, Colab Notebook.

Certifications

- **AWS Academy**, Generative AI Fundamentals *Jun 2026*
- **Infosys SpringBoard**, HTML *Apr 2025*
- **Hackerrank**, Java Basic *Oct 2024*

Achievements

- Secured AIR 7275 in Gate 2026 Conducted by IIT Guwahati
- Received Best Performer Award from Spoorthi Literary Club, LBRCE College.
- Won Second Prize in Hackathon Conducted by LBRCE college.

Positions of Responsibility

- **Student Convener,** Spoorthi Club, LBRCE *Aug 2025 - May 2026*
- **Central Coordinator,** Lakshya 2K26, LBRCE *Dec 2025 - Jan 2026*